Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	9 March 2026
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Upload	Upload microscopic blood smear images through the web interface Support image formats such as JPG, PNG
FR-2	Image Preprocessing	Resize and normalize uploaded images Prepare images for deep learning model input
FR-3	Malaria Detection	Analyse uploaded blood smear images using deep learning models Classify images as **Infected** or **Uninfected**
FR-4	Prediction Result	Display prediction results on the web interface Show confidence score for the prediction
FR-5	Result Visualization	Provide clear visualization of detection results Display results on the user dashboard
FR-6	Result Storage (Optional)	Store detection results for analysis history Maintain prediction records for future reference

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide an easy-to-use interface for healthcare professionals to upload images and view results.
NFR-2	Security	Uploaded medical images and user data must be securely handled to protect patient privacy.
NFR-3	Reliability	The system should produce consistent and accurate predictions for malaria detection.
NFR-4	Performance	The system should process uploaded images and generate results within a few seconds.
NFR-5	Availability	The system should be accessible to users at any time through the web interface.
NFR-6	Scalability	The system should support increasing numbers of users and image uploads through scalable architecture.